"# lagoon-epibenthos-yolo-supplementary"

Lagoon Epibenthos YOLO (lagoon-epibenthos-yolo)

python 3.14.3

YOLO v11

Welcome to the official code repository for the paper:
"Implementing Optimized Computer Vision Algorithm To Underwater Imagery For Identification and Spatial Analysis Of Epibenthic Fauna In Shallow Lagoon Waters"

About the repository

This repository contains the supplementary code, datasets, and model used to detect and analyze epibenthic fauna (*Paranemonia* sp., *Anemonia* sp., and *Brachyura* sp.) in the highly turbid, shallow waters of Logarou Lagoon.

While the custom-trained (for the above 3 classes) YOLOv11s model is provided along with the exported metric files upon the completion of the training, we have also included the ready-to-use extracted data `.xlsx` files so readers can quickly reproduce the statistical tests and validate our findings.

Prerequisites for reviewers

To run these scripts, you will need to install Python and a code editor. We recommend **Visual Studio Code (VS Code)**.

1. Install Python

1. Download **Python 3.14.3** (or newer) from the official Python website. We highly recommend installing the latest stable release for optimal security.
2. Run the installer.
3. **CRITICAL:** During installation, ensure you check the box that says **"Add Python to PATH"** before clicking Install.

2. Install Visual Studio Code (IDE)

1. Download VS Code from code.visualstudio.com.
2. Install the software with default settings.
3. Open VS Code, go to the Extensions tab (`Ctrl+Shift+X`), and install the official **Python extension** (provided by Microsoft).

Installation & Setup

Step 1: Clone or Download the Repository
Download this repository as a `.zip` file and extract it, or clone it via git:

```
git clone https://github.com/yourusername/lagoon-epibenthos-yolo.git
cd lagoon-epibenthos-yolo
```

Step 2: Open the Folder in VS Code
Open VS Code, click `File > Open Folder...` and select the extracted `lagoon-epibenthos-yolo` folder. Open a new terminal in VS Code by clicking `Terminal > New Terminal` from the top menu.

Step 3: Create a Virtual Environment (Recommended)
In the terminal, create and activate a virtual environment to avoid package conflicts:

```
# For Windows:
python -m venv venv
.\venv\Scripts\activate

# For macOS/Linux:
python3 -m venv venv
source venv/bin/activate
```

Step 4: Install Requirements
With your virtual environment activated, install the required packages (Ultralytics, OpenCV, Pandas, SciPy, etc.):

```
pip install -r requirements.txt
```

🚀 Running the Pipeline

All sample data and pre-extracted results are located in the `data-files/` folder.

1. Run the Image Enhancement Script

This script applies the 3-stage radiometric correction (Adaptive Gray-World correction, CLAHE, and Histogram Stretching) to the raw turbid sample images.

```
python scripts/CLAHE_underwater_preprocessor_github_v2.py
```

Run and follow the console instructions.

2. Run the Statistical Analysis

To allow readers to quickly view the study's core results, we provide the ready-made extracted data file (`YOLO_detection_results.xlsx`). This script generates the χ^2 habitat selectivity summaries, size

distribution metrics, and Mann-Whitney U test results.

```
python scripts/unified_statistical_analysis_github_v2.py
```

Again, run and follow the console instructions.

3. Run YOLOv11s Inference (Optional)

If you wish to test the object detection itself, you can run the custom-trained YOLO model on the enhanced images to detect the epibenthic fauna. The custom weights (`best.pt`) are located in the `models/` directory. Inference-ready scripts are provided in the `scripts/For-inference/` directory. Before running open in VScode and adjust properly.

```
python scripts/For-inference/YOLO_on_pics_to_table_and_annot.py
```

📢 Contact & Citation

If you use this code (or logic) in the methodological sequence of your research, please cite our paper (citation details pending publication)! For code-related issues, please open an issue on GitHub.